

IMPORTANT SHORT & LONG QUESTIONS

Chapter 2: Atomic Structure

1. Explain periodic trend of atomic and ionic radius?
2. Quantum numbers complete Topic
3. Which factors affect the ionization energy and mention their trend in period and group?
4. What is Emission Spectrum and how it helps in Electronic Configuration
5. Explain Auf Bau's Principle, Pauli Exclusion Principle and Hund's Rule.
6. How does electronic configuration play an important role in determining the type of semi-conductor?
7. Which factors affect electronegativity and Ionization Energy? OR Write the electronic configuration of a) Br b) Br^- c) Mn d) P e) ^{15}P f) C
8. What is Mass Spectroscopy and its Principle.

Chapter 3: Chemical Bonding

9. Factors affecting Electronegativity.
10. Define dipole moment and their unit?
11. Differentiate between polar and non-polar covalent bond with an example?
12. How we can determine the nature of bond in term of their electronegativity value?
13. Explain the factors on which Bond Enthalpy depends on?
14. Explain peculiar behavior of hydrogen bonding?
15. Calculate bond energy of N_2 , O_2 , F_2 and He? OR Why is the bond energy in F-F lower than in Cl-Cl although the size of F_2 is smaller than Cl_2 ?
16. **VESPR Theory: Identification of the Shapes of Molecules based on Bond and Lone Pairs in Very Important. Memorize the Chart Give in the Book and must learn the Reason for each Shape. MUST do Book Examples + Exercise Problems + Model and Past Papers Questions of this Topic**
17. **Main Points of VBT.** Differentiate between Sigma and Pi Bonds. With examples of H_2 , F_2 , HF, O_2 , N_2 ,
18. **HYBRIDIZATION(sp , sp^2 , sp^3) is Utmost Important Topic (Short + Long). MUST do Book Examples + Exercise Problems + Model and Past Papers Questions of this Topic**
19. Ice is less dense than water, Why?
20. **MOT is most Important. BMO VS AMBO. Draw molecular orbital diagrams of H_2 , He_2 , N_2 and O_2 ?**
21. What is Bond Order, find Bond Order of above diatomic Molecules.
22. Why does Oxygen show paramagnetic nature?

Chapter 4: Stoichiometry

MOST IMPORTANT: Do all Numerical Examples + Exercise Numerical + Numerical in the Model and Past Papers

Chapter 5: States and Phases of Matter

- 23. Postulates of Kinetic Molecular Theory
- 24. **Evaporation and Vapor Pressure (Must do Conceptual Reasoning Based Questions)**
- 25. **Boiling Point and Atmospheric Pressure , do some conceptual Reasoning Based conceptual questions**
- 26. Which factors affect the viscosity of liquid?
- 27. What is the relationship between surface tension and intermolecular force?
- 28. **Crystalline Vs Amorphous Solids and Properties of Crystalline Solids**

Chapter 6: Energetics

- 29. What is heat of reaction? Give two examples of chemical reactions. What are conditions for standard heat of reaction.
- 30. **Explain Hess's Law with example? Or Draw the energy cycle according to Hess's law for: $C + O_2 \rightarrow CO_2$, $\Delta H = -393.51 \text{ kJ/mol}$ If the reaction occurs in two steps: $C + \frac{1}{2}O_2 \rightarrow CO$, $\Delta H = -110.52 \text{ kJ/mol}$ $CO + \frac{1}{2}O_2 \rightarrow CO_2$, $\Delta H = -282.98 \text{ kJ/mol}$**
- 31. **MUST do ALL Examples + Exercise Problems + Numerical of Model & Past Papers**
- 32. **Explain Born Haber Cycle and draw it with the example of formation of NaCl?**
- 33. Which factors affect lattice energy and enthalpy of Hydration?
- 34. How do ionic charge and ionic radius affect enthalpy of hydration?
- 35. Explain briefly the Entropy change of the system.
- 36. **What is meant by Gibbs free energy?**

Chapter 7: Chemical Kinetics

- 37. **Explain Le Chatelier 's principle and how it is important in determining equilibrium state of chemical reaction?**
- 38. **Do all examples , Exercise Numerical, Model and Past Papers Numerical**
- 39. How is activation of energy being important in rate of chemical reaction?
- 40. Explain the effect of catalysts on activation energy of chemical reaction?
- 41. Use Boltzmann distribution curve to explain the effect of temperature and reaction?
- 42. Explain the concept of catalyst and how they increase the rate of reaction by lowering activation energy show their affect by using reaction pathways diagram? Describe the effect of temperature change on rate constant and rate of reaction?

Chapter 8: Equilibrium

- 43. Explain the Law of Mass Action and derive Equilibrium constant K_c expression.**
- 44. Describe the conditions for Equilibrium.
- 45. Do ALL EXAMPLES + Exercise Numerical + Model & Past Papers Numerical**
- 46. Differentiate between homogeneous and heterogeneous equilibria with examples?
- 47. Explain the industrial application of Le Chatelier's principle?
- 48. Synthesis of Ammonia By Haber's Process**
- 49. Synthesis of Sulphuric Acid By Contact Process**

Chapter 9: Acids - Bases

- 50. How we can determine the strength base and acid?
- 51. Identification of Conjugate Acid Base Pairs in Chemical Reaction. Do examples and learn the concept.
- 52. Identification of Lewis Acid Base Pairs in Chemical Reaction. Do examples and learn the concept.
- 53. What is Amphoteric substance. Justify water acts amphoteric in nature.
- 54. How we can calculate the pH of Buffer solutions?
- 55. Explain salt hydrolysis with example?
- 56. Explain the ionization constant of water and calculation of pH, practice relevant examples?
- 57. What is meant by solubility product and precipitation reaction? Construct an expression for K_{sp} ? Or Write the K_{sp} expression for: CoF_2 , MgSO_4
- 58. Explain Common Ion Effect Principle.

Chapter 10: Periodic Table

- 59. Explain the properties of metal, non-metals and metalloids?
- 60. Explain the nature of Reactions of Na and Mg with Water.
- 61. Variation in oxidation number of oxides and chlorides, explain with examples.
- 62. Reaction of Chlorides with water and its effect on PH?
- 63. Briefly explain the thermal stability of hydrogen halides?

Chapter 11: Nitrogen & Sulphur

- 64. What is Photochemical Smog.
- 65. Formation of PAN.
- 66. Explain the uses of sulfur compounds in industry and everyday life, such as in fertilizers, gunpowder and rubber, and in the Synthetic organic chemistry, including the synthesis of dyes, drugs and fragrances.

Chapter 12: Halogens

- 67. Describe the relative reactivity of the halogen elements as oxidizing agents
- 68. Describe the reactions of the halogens with hydrogen and explain their relative reactivity in these reactions.
- 69. Describe the relative thermal stabilities of the hydrogen halides and explain these in terms of bond strengths.**
- 70. Describe the relative reactivity of halide ions as reducing agents.**
- 71. Explain the reactions of halide ions with aqueous silver ions and concentrated sulfuric acid
- 72. Explain the use of chlorine in water purification, including the production of the active species HOCl and ClO⁻ which kill bacteria**

Chapter 13: Environmental Chemistry - Air

- 73. Describe the factors that affect air quality?
- 74. Identify the chemical reactions and processes that occur in the formation of smog and acid rain.

Chapter 13: Environmental Chemistry - Water

- 75. Explain conservation and management strategies for protecting and preserving water resources

Chapters: 15-21 Organic Chemistry

- 76. Explain the shape of ethane molecules in term of structure and bonding?
- 77. Explain the concept of conjugation in alkanes having alternate double bonds?
- 78. Explain the inductive effects of alkyl group on the stability of primary, secondary and tertiary cation formed during electrophilic addition (also explain by Markovnikov addition)?
- 79. Explain the substitution of alcohol: By reaction with HX or KBr, with H₂SO₄ or H₃PO₄, With PCl₃ and heat, With PCl₅, With SOCl₂ OR Write reactions of 1-Propanol with: a) SOCl₂ b) PCl₃ c) PCl₅
- 80. Describe nucleophilic substitution reaction a. reaction with NaOH and heat to produce an alcohol b. reaction with KCN in ethanol and heat to produce nitrile c. reaction with NH₃ in ethanol heated under pressure to produce amine d. reaction with aqueous silver nitrate in ethanol

81. Describe elimination reaction with NaOH in ethanol and heat to produce an alkene as exemplified by bromoethane?
82. Explain the acidity of alcohol compared with water?
83. Explain how you can differentiate between aldehyde and ketone by performing different test?
84. Describe the use of 2,4–dinitrophenylhydrazine Reagent to detect the presence of carbonyl compound?
85. Explain how haloform reaction in primary and secondary alcohol from tertiary alcohol?
86. Describe A. Redox reaction with reactive metal to produce salt and H_2 B. Neutralization reaction with alkali to produce salt and H_2O C. Acid and base reaction with carbonate to produce salt water and carbon dioxide D. Esterification with alcohol
87. Describe the condensation reaction between alcohol and carboxylic acid with conc. H_2SO_4 as a catalyst?
88. Classify amine and describe their structure and chemical properties?
89. Devise multiple step synthetic routes for preparing organic molecules?
90. Explain with equations the reactions of following chlorides with water, also mention pH of resulting mixture. A. $AlCl_3$ b. PCl_3
91. Identify alcohol and carboxylic acid use to prepare methyl propanoate. Give equation for the reaction.
92. When chloroethane reacts with sodium hydroxide different products are formed under different sets of conditions used. Write down the structural formulae of the products which are formed under these Conditions? Which test is used to differentiate between aldehyde and ketone?
93. What is the impact of acid rain on building and human being?
94. Explain the importance of catenation in organic chemistry?
95. Differentiate between molecular formula and empirical formula?
96. What is the characteristics of homologous series?
97. Differentiate between saturated and unsaturated hydrocarbons?
98. Differentiate between electrophile and nucleophile with example?
99. Differentiate between homolytic and heterolytic fusion?
100. Differentiate between hydrolysis and condensation with example?
101. Differentiate between oxidation and reduction reaction?
102. Which type of organic compounds undergo elimination reaction?
103. Explain the phenomena of isomerism with their types?
104. Differentiate between open chain and closed chain hydrocarbon?
105. Which factors determine the reactivity of alkanes? Illustrate the structure and reactivity of ethene molecule?
106. Give mechanisms of hydrohalogenation of alkenes? How we can prepare alcohol from alkene?
107. Prepare epoxyethane from ethene?
108. Prepare methanal from ethene? Classify halogenoalkane into primary, secondary and tertiary alkyl halides?
109. Explain the trend of bond energy of alkyl halides?
110. Give some examples of nucleophilic substitution reaction of alkyl halides?
111. What is meant by elimination reaction meant?
112. Prepare sodium acetate from ethyle acetate?

113. Which test is used to differentiate between primary secondary and tertiary alcohols?
114. Prepare diethyl ether from ethanol?
115. How we can prepare Ester from carboxylic acid?
116. Which type of organic compounds undergo iodoform reaction?
117. Differentiate between aldehyde and ketone?
118. Explain the mechanism of nucleophilic addition of cyanide to an aldehyde?
119. Differentiate between oxidation of aldehyde and ketone with example?
120. What is meant by saponification? Prepare sodium acetate by hydrolysis of Ester?
121. Prepare Quaternary ethyl ammonium iodide by the reaction of alkyl halides with ammonia?
122. Compare basicity of ammonia and methyl amine?
123. How diazonium salts are prepared from primary amine?
124. What type of key reagents are used for synthesis of organic compounds?
125. What type of chemical reaction is given by functional group?
126. Which type of test is used for identification of functional group?
127. Differentiate between monohydric and polyhydric alcohol?
128. Define molar heat of fusion and molar heat of vaporization?
129. Give mechanisms of base catalyzed nucleophilic addition reaction?
130. Give mechanisms of acid catalyzed nucleophilic addition reaction with example? Explain with the help of equations for acid / base behavior of oxide and the hydroxide NaOH, $Mg(OH)_2$ including where relevant amphoteric behavior in reaction with acid and base?
131. Explain the use of chlorine in water purification?
132. Describe the factor affecting air quality?

LONG QUESTIONS

1. Mentioned all the reaction that are used to prepare alcohol?
2. Illustrate the reaction involving H-atom of the carboxyl group?
3. Explain the reaction involving the OH group of carboxylic acid?
4. How we can prepare amine by the reduction of nitrogen containing functional group?
5. Give mechanisms of reaction that involved in preparation of alkyl halides?
6. Explain the steps involved in free radical substitution of ethane?
7. Practice nomenclature of alkane, alkene, alkyne and carboxylic acid and alcohol?
8. Draw a table to represent functional group related to organic compound?
9. Explain conservation and management strategies for protecting and preserving water resources?
10. Describe the structural isomerism and classify it into their types? Practice some example to identify different types of structural isomerism?
11. State Markovnikov's rule. Write the mechanism and identify the major product when propene reacts with HBr.
12. Illustrate all the method used for preparation of amine?
13. Write down the reagent and conditions used to prepare propanoic acid from:
14. Propanol
15. Propane nitrile
16. Methyl propanoate

17. Consider an alcohol with molecular formula $C_4H_{10}O$. It exists in different isomer form: a. Write structural formula of four isomers and write their name. b. What type of isomerism is shown by these compound?

Important Mechanisms Must Learn and Practice

- Describe the mechanism of free radical substitution in alkanes exemplified by methane and ethane.
- Describe the mechanism of electrophilic addition in alkenes, using bromine / ethene and hydrogen bromide /propene as examples
- Describe the mechanism of the nucleophilic reactions of hydrogen cyanide with aldehydes and ketones
- Mechanisms of free-radical substitution, electrophilic addition, nucleophilic substitution, nucleophilic addition, electrophilic addition and elimination reactions.